METRIC

MIL-P-63245B (AR)
16 October 1995
SUPERSEDING
DOD-P-63245A (AR)
28 September 1981

MILITARY SPECIFICATION

PROJECTILE, HE, 155MM, M795
METAL PARTS ASSEMBLY

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the manufacturing and assembly of the metal parts for Projectile, HE, 155MM, M795 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL (Advisory)

TT-C-490 - Cleaning Methods and Pre-treatment of Ferrous Surfaces for Original Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FCS 1320
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MILITARY (Mandatory)

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-R-13562 - Rotating Bands, Welded Overlay, Ferrous and Nonferrous, for Projectiles

STANDARDS

FEDERAL (Advisory)

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing

MILITARY (Advisory)

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-1949 - Inspection, Magnetic Particle

MILITARY (Mandatory)

MIL-STD-1168 - Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9326055 - Projectile, 155MM, HE, M795 Metal Parts Assembly

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12977200 - Body Assembly

12944446 - Packing and Marking, 155mm, Metal Parts Assembly Projectile Pallet

9329574 - Packing and Marking for Pallet Wood, Ammunition for Projectile, 155MM, HE M795

INSPECTION EQUIPMENT DRAWINGS

9280336 - Envelope Drawing, Ultrasonic Inspection Equipment and Procedure for Body, Machined, 155MM, M795

9280308 - Ultrasonic Inspection Standard, Longitudinal and Radial Cracks, M795, Body

9280309 - Ultrasonic Inspection Standard Circumferential and Piping Defects 155MM, M795

9280441 - Envelope Drawing, Magnetic Particle Inspection, Band Area, 155MM M795

(Copies of specifications, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the Contracting activity or as directed by Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E8 - Tension Testing of Metallic Materials
E10 - Brinell Hardness of Metallic Materials
E103 - Rapid Indentation Hardness Testing of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text

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of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Projectile metal parts assembly. The projectile assembly and components shall comply with all requirements specified on Drawing (dwg.) 9326055 and with all the requirements specified in applicable specifications.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied shall be uniform throughout each lot of projectiles and shall be so performed and controlled as to produce in the projectile the mechanical properties specified on the applicable drawing. No projectile shall be reheat treated if it fails to meet the applicable drawing requirements without prior Government approval.

3.4 Hydrostatic test. The projectile shall withstand an internal hydrostatic pressure in accordance with requirements without evidence of leakage, cracks, ruptures or permanent distortion.

3.5 Performance. There shall be no break-up or separation of metal parts in the gun bore or in flight.

3.6 Projectile material defects.

3.6.1 Projectile body. The projectile body, including the cavity shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects.

3.6.2 Projectile cavity. In addition to the requirements of 3.6.1 the projectile cavity shall be free from scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and all surface discontinuities which have a radius smaller than 0.76 millimeters(.03 in.).

3.6.3 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first months production. However, establishment of visual standards will not waive the requirements for compliance with 3.6.1 and 3.6.2 (see 6.5).

3.7 Rotating band.

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3.7.1 Rotating band application. The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process, in this instance, is defined as the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary wire feed system. The procedures for maintaining, controlling, and monitoring the process must be submitted to the technical agency for approval prior to the start of production. Documented records of the process control must be made available to the Government upon request (see 6.13).

3.7.1.1 Minimum melting of steel. Upon removal of the rotating band material, the steel band seat shall demonstrate visual evidence of minimum melting due to gilding metal adhesion during application of the rotating band. Visual comparison standards will be provided by the Government.

3.7.1.2 Subsurface defects. There shall be no evidence of either intergranular penetration of gilding metal or weld start-stop cratering when the body band seat is turned down to 147.19 plus or minus 0.25 millimeters (mm) diameter and tested as specified in 4.4.3.2.

3.7.2 Process control. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface. A process requirement that will control this feature is depicted on drawing 12977199. The contractor may, at its option, use a control other than the one defined on the drawing. The manufacturing procedure and the inspection plan for controlling process shall be forwarded to the responsible technical agency for approval prior to its implementation into production.

3.7.3 Foreign matter on band seat. Prior to banding, the band seat shall be free from any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.8 Metal defects

3.8.1 Components or assemblies. All components or assemblies shall be free from all metal defects, in accordance with the defect standards (see 3.6.3), including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits (see 3.6.3).

3.8.2 Rotating band. The rotating band and band seat areas shall have no discontinuities, cracks, lack of bonding, or other metal defects in accordance with defect standards (see 3.6.3).

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3.8.3 Body. Projectiles shall be free of any discontinuities equal to or in excess of the ultrasonic inspection criteria for the respective zones. The inspection criteria are notches and holes in standards (see 9280308 and 9280309) which simulate larger actual cracks and piping flaws.

3.9 Interior cavity finish. The interior cavity after painting shall be free from bare spots, blisters, grease, dirt and other foreign matter. There shall be no pool of paint in the bottom of the cavity.

3.10 Slow bend precracked charpy test. The minimum crack strength shall be 25 KSI $\sqrt{in}$. (thousand pounds per square inch square root inch)

3.11 Painting. All paint shall be dry to the touch before packing for shipment.

3.12 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.13 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize

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submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Inspection and tests (non ballistic).

4.3.1.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Projectile, 155MM, HE, M795 Metal Parts Assembly	9326055	45
Body Assembly	12977200	25
Body Machined	12977199	25
Body Machined (Prior to cutting Thread and Band Seat)	12977199	5
Base Cover	9326051	5

All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, materials, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to

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significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic tests.

4.3.2.1.1 Submission. Twenty complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and subsequent ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1) and shall consist of twenty assemblies.

4.3.2.1.2 Tests to be performed and acceptance criteria. Performance tests will be conducted on twenty (20) assemblies in accordance with 4.4.5.1., utilizing the procedures of 4.5.9 to determine compliance with 3.5.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I First Article Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Assemblies and Components	SHEET 1 OF 2	DRAWING NUMBER NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile, 155M, HE, M795 Metal Parts Assembly			
	(Dwg. 9326055)			
	Examination of characteristics	25	3.2	4.4.2.6
	Ballistic Performance for structural intergrity	20	3.5	4.5.7
	Body Assembly			
	(Dwg. 12977200)			
	Examination of characteristics	25	3.2	4.4.2.5
	Band chemical analysis	5	3.2	4.5.3.4
	Properties of rotating band weld	5	3.2	4.4.3.2/4.5.4
	Hydrostatic test	25	3.4	4.5.1
	Ultrasonic inspection	25	3.8	4.5.5
	Body, Machined			
	(Dwg. 12977199)			
	Examination for Defects	25	3.2	4.4.2.4
	Yield strength and elongation test	2	3.3	4.5.3/4.4.3.1
	Slow bend precrack Charpy	1	3.10	4.5.3.3/4.4.3.1.4

NOTES:

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TABLE I First Article Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Assemblies and Components	SHEET 2 OF 2	DRAWING NUMBER NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base Cover (Dwg. 9326051)			
	Examination of characteristics	5	3.2	4.4.2.1
	Maximum temperature of mult	Note A	3.2	4.4.3.6

NOTES:

A. One per hour of furnace operation

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the formation provisions of MIL-A-48078 and the following:

4.4.1.1 Heat treatment inspection lot. Heat treatment inspection lots shall not exceed one mill heat of steel or 2000 components, whichever is smallest. Whenever a break in heat treating of eight (8) hours or more occurs or a furnace is shut down for any length of time, the heat treatment inspection lot will be terminated and a new lot shall start. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into larger heat treatment inspection lots after qualifying with individual batches provided qualifying procedures are submitted to and approved by the Government and provided all other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels cited in the Classification of Characteristics paragraphs.

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TABLE II 0-1 ATTRIBUTE PLANS

Lot Size	INSPECTION LEVELS I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10

Numbers under inspection levels indicate sample size; asterisk indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system as applicable.

c. Tool control dimensions. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to

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the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

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Quality Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63245B (AR)

PARAGRAPH 4.4.2.1	TITLE Base Cover	SHEET 1 OF 1	DRAWING NUMBER 9326051
			NEXT HIGHER ASSEMBLY 12977200

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined.			
Major 101	Thickness	Level III	3.2	Gage
Minor	None Defined			

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-63245B (AR)

PARAGRAPH 4.4.2.2	TITLE Mult	SHEET 1 OF 1	DRAWING NUMBER 9345132
			NEXT HIGHER ASSEMBLY 12977199

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined.			
Major 101	Temperature of mult, max.	One per hour	3.2	4.4.3.5
Minor	None Defined.			

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-63245B (AR)

PARAGRAPH 4.4.2.3	TITLE Projectile, Prior to Nosing	SHEET 1 OF 1	DRAWING NUMBER 12977199
			NEXT HIGHER ASSEMBLY 12977200

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined.			
Special 10	Absence of material defects	100%	3.6.2	4.5.2/Visual
Major	None Defined.			
Minor	None Defined			

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.4	TITLE Body Machined	SHEET 1 OF 3	DRAWING NUMBER 12977199 / NEXT HIGHER ASSEMBLY 12977200

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
10	Projectile cavity free of defects	100%	3.6	4.5.2/Visual
12	Precracked Charpy test	4.4.3.1.3	3.10	4.5.3.3
Major				
101	Diameter of rotating band seat (*)	Level III	3.2	Gage
102	Runout of rotating band seat (*)	Level III	3.2	Gage
103	Location of rotating band seat (*)	Level III	3.2	Gage
104	Width of rotating band seat (*)	Level III	3.2	Gage
105	Cleanliness of rotating band seat (*)	Level III	3.2	Visual
106	Yield strength and elongation of body	4.4.3.1	3.2	4.5.3
107	Thickness through the base	Level III	3.2	Gage
108	Radii or chamfers present	Level III	3.2	Visual
109	Runout of pitch diameter of thread with bourrelet and front face	Level III	3.2	Gage
110	Runout of ogive with bourrelet (three locations)	Level III	3.2	Gage

NOTES:

* Prior to banding

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.4	TITLE Body Machined	SHEET 2 OF 3	DRAWING NUMBER 12977199
			NEXT HIGHER ASSEMBLY 12977200

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
111	Runout of bourrelet relief with bourrelet	Level III	3.2	Gage
112	Runout of boattail with bourrelet	Level III	3.2	Gage
113	Diameter of bourrelet, min.	Level III	3.2	Gage
114	Diameter of bourrelet relief	Level III	3.2	Gage
115	Length from front face to first basic diameter	Level III	3.2	Gage
116	Length from front face to second basic diameter	Level III	3.2	Gage
117	Diameter at 186.34mm from forward end	Level III	3.2	Gage
118	Diameter at 313mm from forward end	Level III	3.2	Gage
119	Length from rear face to forward basic diameter on boattail	Level III	3.2	Gage
120	Length from rear face to rear basic diameter on boattail	Level III	3.2	Gage
121	Length from front face to relief of ogive	Level III	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-63245B (AR)

PARAGRAPH 4.4.2.4	TITLE Body Machined	SHEET 3 OF 3	DRAWING NUMBER 12977199
			NEXT HIGHER ASSEMBLY 12977200

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
122	Length from rear face to bottom of cavity	Level III	3.2	Gage
123	Length from front face to rear of bourrelet relief	Level III	3.2	Gage
124	Length from front face to forward face of bourrelet relief	Level III	3.2	Gage
125	Thickness of body wall in ogive area, min. (2 locations)	Level III	3.2	Gage
126	Variation of wall thickness in ogive area	Level III	3.2	Gage
127	Variation of wall thickness in bourrelet relief	Level III	3.2	Gage (Note a)
128	Pitch diameter of threads, max.	Level III	3.2	Gage
129	Minor diameter of thread, max.	Level III	3.2	Gage
130	Length of effective thread, min.	Level III	3.2	Gage
131	Length, total	Level III	3.2	Gage
132	Runout of boattail cavity	Level III	3.2	Gage
133	Runout aft of band seat	Level III	3.2	Gage
134	Diameter aft of band seat	Level III	3.2	Gage
MINOR	None defined			

NOTES:

a. Wall variation will be checked in four locations of bourrelet relief, approximately 25 millimeters between measurements.

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Quality Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63245B (AR)

PARAGRAPH 4.4.2.5	TITLE Body Assembly	SHEET 1 OF 2	DRAWING NUMBER 12977200
			NEXT HIGHER ASSEMBLY 9326055

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1	Hydrostatic Test	100%	3.4	4.5.1
SPECIAL				
10	Absence of metal defects	100%	3.6	Visual
11	Presence of base plate, unbroken base plate weld and unperforated base plate	100%	3.2	Visual
12	Hydrostatic test stamp missing	100%	3.2	Visual
13	Ultrasonic test	100%	3.8	4.5.6
14	Ultrasonic test stamp missing	100%	3.2	Visual
15	Magnetic particle test (band area)	(Note a)	3.2	4.5.4
MAJOR				
101	Diameter of rotating band ribs	Level III	3.2	Gage
102	Runout of rotating band with bourrelet	Level III	3.2	Gage
103	Diameter of obturator band seat	Level III	3.2	Gage
104	Width of obturator band seat	Level III	3.2	Gage
105	Location of cannelure (3 locations)	Level III	3.2	Gage
106	Diameter of rotating band cannelure (3 locations)	Level III	3.2	Gage

NOTES:

a. One sample per 8 hours per band machine. Failure shall cause rejection of assys represented.
b. Same sample(s) selected for magnetic particle shall be utilized.

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Quality Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.5	TITLE Body Assembly	SHEET 2 OF 2	DRAWING NUMBER 12977200
			NEXT HIGHER ASSEMBLY 9326055

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
107	Location of rotating band	Level III	3.2	Gage
108	Properties of rotating band weld	(Note b)	3.7	4.4.3.2
109	Band chemical analysis	4.5.3.4	3.2	Gage
110	Location of basic diameter (Rotating band)	Level III	3.2	Gage
111	Absence of suface voids, seams, iron deposits or cracks in rotating band	Level III	3.2	Visual/Probe/Scale
112	Length, total, max.	Level III	3.2	Gage
113	Weight	Level III	3.2	Gage
Minor	None Defined			

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Projectile, HE, 155MM, M795 Metal Parts Assembly	SHEET 1 OF 1	9326055
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1	Ballistic perfomance test for structural intergrity	4.4.5	3.5	4.5.7
SPECIAL				
10	Pool of Paint in cavity	100%	3.9	Visual-Probe light
20	Absence of metal defects	100%	3.6	Visual-Probe light
MAJOR				
101	Diameter of bourrelet, max.	100%	3.2	Gage
102	Rotating band undamaged	Level III	3.2	Visual
103	Cleanliness of cavity	Level III	3.2	Visual
104	Salt Spray Test	4.4.3.4	3.2	4.5.6.1
105	Paint on threads	Level III	3.2	Visual
106	Absence of burrs	Level III	3.12	Visual
107	Protective coating thickness	4.4.3.4	3.2	4.5.6.2
Minor	None Defined			

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4.4.3 Testing.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness. Each final heat treated projectile from a heat treated lot shall be identified and subjected to a hardness test on the surface indicated. One (1) hardness reading shall be taken. The hardness reading obtained shall be used as the basis for selecting the tension test samples of paragraph 4.4.3.1.2.

4.4.3.1.2 Yield strength and elongation. Two final heat treated projectiles from a heat treated inspection lot shall be selected for the tension test. The projectiles selected shall be those which have the highest and the lowest hardness reading from 4.4.3.1.1. From each of these projectiles three specimens shall be tested in accordance with 4.5.3.

4.4.3.1.2.1 Retest. In the event of rejection, new samples of higher or lower hardness values or both, as applicable, may be selected and tested in order to establish an acceptable hardness range. Once this range has been established, all projectiles whose hardness values fall outside the range shall be rejected or subjected to retemper or heat treat cycle if approved by the Government. An acceptable hardness range is one which will provide acceptable yield and elongation results.

4.4.3.1.3 Slow bend precracked charpy test. Beginning with the first heat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated projectile from each heat treated lot shall be subjected to the precrack charpy test. The projectile selected shall be of the highest hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested. No individual fracture toughness shall be less than 25 KSI $\sqrt{in}$. If one test is invalid, acceptance may be based on the fracture toughness of the valid sample. If either individual fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to retemper and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be reinstituted, until three consecutive heat treated lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

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c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government.

4.4.3.2 Inspection of band seat.

4.4.3.2.1 Rotating band seat location and runout. The contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.4.2.2. When the contractor establishes a control method, the manufacturing procedure, and inspection plan shall be forwarded to the Technical Agency for approval prior to use in accordance with 3.7.2. No change to these controls will be made once established without approval from the Technical Agency.

4.4.3.2.2 Welder qualification sampling. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the test outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat metal (steel) (see 6.12).

These bodies will be subjected to the following tests:

a. Visual examination for compliance with 3.7.1.1 minimum melt requirements.

b. Visual examination for compliance with 3.8.2 metal defects, using the technique described in 4.5.4.

c. Turn down band seat to the diameter specified in 3.7.1.2 and again visually inspect in accordance with 4.5.4. Any defect indication per 3.7.1.2 at this depth is considered to be a reject.

Should any sample for any of the above tests fail to comply with minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.4.3.2.3 Post Qualification Sampling.

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4.4.3.2.3.1 Etched bandseat examination. Reduced sampling of one in each shift (not to exceed ten hours) of production, or fraction thereof, per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable (see 6.13). One body per welder will be randomly selected and inspected according to the following criteria:

a. Visual examination for compliance with 3.7.1.1, minimum melting. The body selected for this test need not be the test sample having the lower rotating band iron content when tested in accordance with 4.5.3.4.

b. Visual inspection for compliance with 3.8.2, metal defects, using the technique described in 4.5.4. If the sample fails to comply with 3.8.2, then the band seat will be turned down to the diameter specified in 3.7.1.2 and again visually inspected in accordance with 4.5.4.

c. Should any sample fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

d. Two consecutive failures when tested for 4.4.3.2.2.1a, or two consecutive failures when tested for 4.4.3.2.2.1b, for any welding machine will mandate the return to the original welder qualification sampling of 4.4.3.2.1 However, reduced sampling of one in each shift of production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.4.3.2.3.2 Sub-surface defect examination. The body having the highest acceptable iron content rotating band, when tested per 4.5.3.4 during each shift of post qualification sampling, shall be tested for compliance with 3.7.1.2. This body will have its band seat turned down to the diameter specified in 3.7.1.2 and visually inspected per 4.5.4. Failure of the body to comply with the specified requirements will require rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body will also require that the body exhibiting the highest iron content from each welder will be similarly tested. Failure of any body will result in rejection of the quantity from each discrete welder represented by the sample. Two consecutive failures on any welding machine will mandate the return to the original welder qualification sampling of 4.4.3.2.1. However, reduced sampling of one in each shift of production, or fraction thereof, may be implemented after the first five banded bodies have been found acceptable.

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4.4.3.3 Salt spray. After qualification in accordance with TT-C-490 or commercial equivalent as approved by the contracting officer, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.4 Protective coating - thickness. Two samples shall be selected at random to represent each shift or fraction thereof, of continuous paint production. The sample projectiles shall be inspected for paint coating thickness at a minimum of four separate locations on each projectile. Any projectile failing to meet the applicable requirements shall be classed defective and every projectile painted since the last acceptable sample shall be rejected.

4.4.3.5 Mult temperature. One mult shall be checked for maximum temperature during each hour, or fraction thereof, of furnace operation. The inspection point and method shall require approval of the Government technical agency. If any mult checked exceeds the applicable requirement, all mults in that heat treatment batch shall be rejected.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.4.5 Ballistic performance test for structural integrity. Testing shall be performed as specified in 4.5.7.

4.4.5.1 Initial inspection. Beginning with the first lot produced and continuing until three consecutive lots have complied with requirements specified, 20 sample projectiles shall be randomly selected for this test, 20 sample projectiles shall be observed for structural integrity of parts when tested in accordance with 4.5.7. If any sample projectiles fail to comply with the requirements for structural integrity of parts, the lot shall be rejected.

4.4.5.2 Reduced inspection. After three consecutive lots have complied with 4.4.5.1, 10 sample projectiles shall be randomly selected from each lot for this test. The 10 sample projectiles shall be observed for structural integrity of parts. If any of the sample projectiles fail to comply with the requirement for structural integrity of parts, the lot shall be rejected. If the lot is rejected the inspection criteria reverts back to paragraph 4.4.5.1.

4.5 Methods of inspection.

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4.5.1 Hydrostatic test. The unpainted projectile shall be mounted in a suitable test fixture and be subjected to the pressure specified on the applicable drawing. Any projectile which fails to comply with the applicable requirements shall be classed defective and removed from the lot.

4.5.2 Visual examination. The internal cavity of the projectile shall be inspected visually prior to nosing using probe lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.6.3 shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements of 3.6 shall be classed defective and removed from the lot.

4.5.3 Mechanical properties.

4.5.3.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area designated by the drawing to be hardness tested shall be prepared by machining to a sufficient depth, 0.38 millimeters (.015 in.) minimum, to remove scale and get below the decarburization zone. The preparation process to be used shall require Government approval. Manual grinding shall not be used. The surface preparation requirement may be eliminated if the following conditions are met:

(1) Lab results confirm that current production process produces a decarburized zone of 0.076 millimeters (.003 in.) maximum in the test area. The contractor shall provide samples, or photo-micrographs, of high and low hardness samples to the Government for approval. Samples may be the same as for 4.5.3.2 below.

(2) A controlled atmosphere is maintained in the austenitizing furnace.

(3) Depth of decarburization shall be reverified whenever precrack charpy testing is required (see 4.4.3.1.4), but only for the first heat treat lot.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 10mm Brinell ball. The body shall be securely supported so that no rocking or shifting

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of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000kg. The amount of penetration of the indentor (10mm ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. One hardness reading shall be taken.

4.5.3.2 Yield strength and elongation. The two projectiles selected for testing shall have three specimens taken from each and tested as follows:

a. Base section. The base section tensile specimen shall be taken transversely. Failure to meet the yield strength or elongation requirements shall require a retest from the remaining portion of the base. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot.

b. Side wall. Two longitudinal specimens shall be taken 180° from the sidewall. Location shall be centered under the rotating band seat and in the middle of the wall thickness in that area. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

Testing shall be conducted in accordance with American Society for Testing and Materials, Method E8 with the largest substandard size round specimen possible.

4.5.3.3 Slow bend precrack charpy. The two specimens shall be taken from locations C1 and C2 on Figure 1. The test shall be conducted at a specimen temperature of - 54°C (-65 degrees Fahrenheit) . The test shall be conducted in accordance with the procedure defined in the Appendix of this specification with the notches machined across the rear face of the specimen. Locations C3 and C4 on Figure 1 shall be used for replacement specimens for locations C1 and C2 respectively.

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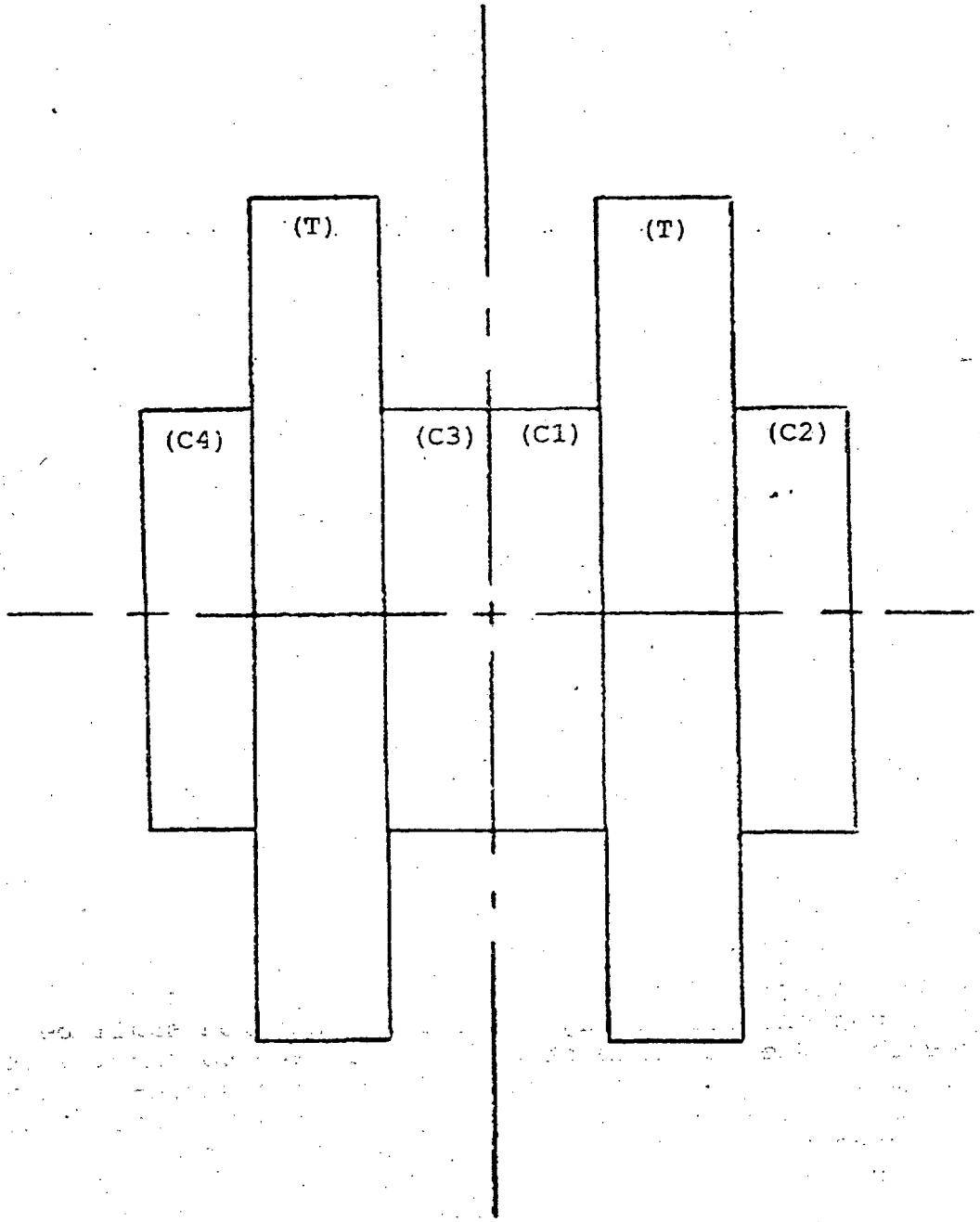


FIGURE 1.

APPROXIMATE LOCATION OF TENSION AND CHARPY BASE SPECIMENS
(T = TENSILE, C = CHARPY)

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4.5.3.4 Band chemical analysis. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been tested and found acceptable. Reduced sampling of one in each half-shift of production, or fraction thereof, per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been tested and found acceptable. Sample chips shall be obtained from each band from the last 12.7 mm removed when machining to the finished diameter. All chips shall be collected and identified with the body from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be analyzed by any method that has been approved by the technical agency. The sample may not be cleaned with a magnet. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

4.5.4 Magnetic particle inspection (Rotating band area). The inspection shall be in accordance with MIL-STD-1949, or equivalent commercial standard as approved by the contracting officer, using the continuous, wet, fluorescent method. The suspension shall be applied to the machined area beneath the rotating band. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 milliliters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. After draining the excess suspension, the rotating band area shall be inspected under ultra-violet light for the presence of defects. The adequacy of the inspection process shall be checked at the start of the daily production and at eight-hour intervals thereafter using a standard in accordance with dwg. 9280441. The plug in the standard shall be clearly defined for a full 360 degrees on the outside of the body. If the inspection process is determined to be inadequate, the condition shall be corrected and the proceeding 8 hours' production shall be reinspected. The requirement of MIL-STD-1949 to check system effectiveness by means of steel ring specimen shall be waived. The provisions for the removal of additional material shall not apply to the inspection beneath the rotating band on the samples selected in accordance with 4.4.3.2.

4.5.4.1 Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I, or equivalent commercial standard as approved by the contracting officer. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

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4.5.5 Ultrasonic inspection. The projectiles shall be ultrasonically inspected using equipment and procedures in accordance with 4.5.5.1. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards, in accordance with 4.5.5.3, shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.8 shall be classed defective and removed from the lot. All projectiles rejected by ultrasonics shall be visually inspected for the location of the discontinuity. If the discontinuity can be observed visually, it shall be characterized as a critical defect, and appropriate action shall be taken.

4.5.5.1 Equipment and procedures. The ultrasonic inspection shall be performed with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Should a stream or thin film of liquid be used to couple this energy, controls shall be used to monitor the coupling and assure its effectiveness. Such monitoring is not required if submergence in a liquid is used.

Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the wall and bourrelet ultrasonic regions, ultrasonic energy shall move within the walls of the projectile with separate significant axial or circumferential components of velocity to detect flaws located on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential.

Criteria for detection of cracks are machined notches which simulate actual larger cracks. Details are contained on drawings 9280308 and 9280309.

All notches machined on the standards are targets for rejection. That is, any projectile that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected. Relief portions of radially oriented notches where the full depth approaches the surface are not inspection criteria.

4.5.5.2 Calibration and operation procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum, the adequacy of the inspection process shall be checked by using the standard at the start and finish of each production shift or change in operator, as well as at the shift midpoint during production. All channels shall properly indicate a rejection signal when encountering the notches mentioned in 3.8. If the inspection process is determined

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to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which is a minimum, documents the use of standards, transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.5.3 Ultrasonic standards. The ultrasonic inspection standard shall be provided by the contractor for use during the ultrasonic inspection. The standards shall be machined from an acceptable projectile in accordance with drawings 9280308 and 9280309 and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standard is required

4.5.5.4 Operator qualification. Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 or equivalent commercial standard as approved by the contracting officer, with the following exception:

Each facility shall have as a minimum an individual who qualified as a Level II inspector in all respects but need not include experience.

4.5.6 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon.

4.5.6.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 or commercial equivalent as approved by the contracting officer, with the exception that test samples shall be selected in accordance with 4.4.5.1.

4.5.6.2 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially available equipment.

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4.5.7 Ballistic performance test for structural integrity. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample projectiles shall be inert loaded, assembled and fired at ambient temperature in a howitzer using a PXR6297A1 propelling charge. The propelling charges shall be uniformly temperature conditibned to provide a chamber pressure of 399.6 Mpa + 27.6 Mpa (58,000 + 4000 PSI). Photographic and instrumental coverage shall be provided to assure observation of round at muzzle exit and over the predicted trajectory. Range and deflection shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts.

5. PACKAGING

5.1 Packing.

5.1.1 Packing - Level A. The projectile shall be palletized in accordance with drawing 12944446.

5.1.2 Alternative Packing - Level A. Only with the specific approval of the contracting officer, shall the projectile be palletized in accordance with drawing 9329574.

5.2 Marking. Markings shall be in accordance with drawing 12944446 or 9329574 as applicable.

5.3 Lot numbering. Lots shall be numbered in accordance with MIL-STD-1168.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile covered by this specification is intended for use on the Projectile, 155MM, HE, M795. This specification contains requirements suitable for production.

6.2 Acquisition requirements. See MIL-A-48078 (AR).

6.3 Submission of inspection procedures and equipment designs for approval. (See MIL-A-48078) Submit equipment designs, as required, to Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contractor Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.6 Ballistic Performance test summary.

Test	Sample size (see 4.4.5.1)	Requirements
Struct. integrity	*20 10	No breakup or metal parts separation (see 3.5)

* Applicable only for First Article Test and three consecutive fully compliant lots.

6.7 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, ARDEC, Picatinny Arsenal, NJ 07806-5000, ATTN: AMSTA-AR-QAA-A and AMSTA-AR-FSI-M.

6.8 Visual standards. Visual standards will be established at contractors plant in compliance with 3.6.3.

6.9 Corrosion preventive compounds. Materials complying with specification MIL-C-16173, Grade 5, have been found to be compatible with TNT.

6.10 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirements	Applicable DID
4.5.2	Test Report	DI-T-1906
4.5.3	Test Report	DI-T-1906

(Copies of data item descriptions required by the contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.11 Process notes. The following notes are provided for information only:

a. Heat treat - Austenitize - 802 °C to 843 °C (1475 °F to 1550 °F) for one hour minimum.

- Oil Quench - 43 °C to 60 °C (110 °F to 140 °F)
- Temper - 560 °C (1040 °F) min. for 1 hr. min.

b. Cooling of forgings should not be retarded. Slow cooling can cause the formation of carbides.

c. Spheroidizing after forging can be beneficial in rough machining, nosing and heat treatment.

d. Nosing should be performed in a minimum of two steps.

e. Quenching over spuds is recommended to assure adequate quenching of the base.

f. The hydrotest plug should be designed with a one quarter inch, minimum thread relief to prevent loading of initial threads on the projectile.

g. Examination should be made of all hydrotest plugs after a projectile is blown during testing. Badly nicked or damaged plugs may cause destruction of otherwise acceptable items.

6.12 Suggested etching procedures for removing gilding metal overlay .

a. Recommended. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.25 to 0.38 mm thick will dissolve in 10 to 14 hours. Renew solution as necessary.

b. Solution - 500 grams Chromic oxide (Cr_2O_3) (commercial flakes), 50 ml. sulphuric acid (H_2SO_4) (technical grade), 950 ml. water (H_2O) (distilled).

This solution may be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.25 to 0.38 mm thick will require approximately two to three hours to dissolve in a fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

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6.13 Band welding. It is recommended that the welding operation of the overlay welded band be guided by Frankford Arsenal "Process Recommendations for Banding 155MM, M483 Projectile" dated 19 April 1973. Requests for copies should be submitted to: Commander, ARDEC, ATTN: AMSTA-AR-CCH, Picatinny Arsenal, NJ 07806-5000.

6.14 Subject term (keyword) listing.

Hydrostatic test
Reduced inspection
Salt spray
Slow bend precracked Charpy test
Ultrasonic test

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-0009)

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APPENDIX

A PRECRACKED CHARPY FRACTURE TOUGHNESS METHOD FOR HF-1 STEEL

10. SCOPE

10.1 Scope. This test method covers the determination of a fracture toughness, K_q, of a Charpy-type specimen (see Fig. 2) containing a fatigue crack tested in slow bending. This test method will be applicable for HF-1 steel only.

This fracture toughness will be sensitive to changes in the plane-strain fracture toughness providing the strength of the specimen is determined primarily by crack propagation and not by plastic instability in the Charpy-type specimen.

20. APPLICABLE DOCUMENTS

20.1 ASTM Standards:

E-4 Practices for Load Verification of Testing Machines
E-8 Methods of Tension Testing of Metallic Materials
E-23 Methods for Notched Bar Impact Testing of Metallic Materials
E-139 Practice for Conducting Creep, Creep-Rupture, and Stress-Rupture Tests of Metallic Materials
E-399 Test Method for Plane-Strain Fracture Toughness of Metallic Materials
E-812 Crack Strength of Slow-Bend Precracked Charpy Specimens of High-Strength Metallic Materials

30. SUMMARY OF METHOD

30.1 Summary of method. This test method employs a Charpy specimen provided with a sharp notch terminating in a fatigue crack tested in three-point bending using fixtures that minimize the contribution of friction forces to the measured applied load. The maximum load in the test is recorded and the fracture toughness is determined from this value and the original dimensions of the specimen.

40. PRECAUTIONARY NOTES

40.1 Precautionary notes. When preparing specimens and testing for fracture toughness, it is important that certain precautions be kept in mind.

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40.1.1 Toughness. The fracture toughness of some materials will be sensitive to the rate of loading and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The fracture toughness will be influenced by the testing temperature and calibration relations may change with testing temperature.

50. DEFINITIONS

50.1 Fracture toughness, K_q. As determined by this method, the fracture toughness is defined as the nominal stress intensity in a notched and cracked bend specimen calculated on the basis of the length of the initial uncracked ligament and the maximum load sustained in the test.

50.2 The following symbols are used in this test method:

a - crack depth (machined notch plus fatigue crack) (in.)
B - specimen thickness (in.)
W - specimen width (depth) (in.)
f(a/W) - power series of ratio (a/W) from ASTM E-812
P_{max} - maximum load applied (lbf)
S - span between support rollers (in.)
K_f - maximum stress intensity factor during fatigue (KSI $\sqrt{in}$).
K_q - fracture toughness obtained in the test (KSI $\sqrt{in}$.)
E - Young's modulus of elasticity (KSI)
T - test temperature (°F)

60. APPARATUS

60.1 Tension-testing machine. Equipment shall conform to the requirements of Practices ASTM E-4.

60.2 Bend-test fixture. The procedure involves testing of the specimen in three-point bending. The bend test fixture should provide support rollers that are free to rotate and move apart slightly during loading (for example, see the Test Method E-399 bend test fixture for general principles of design). The nominal span between support rollers shall be 1.57 in. (between 1.50 in. and 1.65 in.). If the diameter of the support rollers is between 0.30 in. and 0.39 in. there will be negligible change in span during deformation of the specimen.

60.3 Temperature control. The testing temperatures shall be controlled by use of an integral temperature conditioning chamber as part of the testing apparatus. Any suitable method may be used to temperature condition the specimen providing the region including the notched and cracked section can be uniformly maintained within ±2°F of the desired test temperature.

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60.4 Temperature measurement. The temperature of the specimen during testing shall be measured by means of two thermocouples attached on either side of the notched and cracked section. These thermocouples may be spot welded in place; however, the spot weld locations shall avoid the crack path. At the test temperature, the difference between the temperatures indicated by these two thermocouples shall not exceed 1°F and the average of these two temperatures shall be within ±2°F of the desired test temperature.

60.4.1 Thermocouples. Temperatures shall be measured with calibrated thermocouples used in conjunction with potentiometers or millivoltmeters. Such measurements are subject to various errors; reference should be made to ASTM E-139 for a discussion of errors.

60.4.2 Calibration. The temperature measuring apparatus shall be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that the accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS, AND PREPARATION

70.1 Dimensions. The precracked Charpy bend specimen is shown in Fig. 2. The crack depth, a, (starter notch depth + fatigue crack length) is nominally equal to one half the specimen width, W, and shall lie within the range between 0.45 and 0.55 W.

70.1.1 Crack starter. The crack starter shall consist of a V-notch having an included angle from 45° to 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. is usually necessary in order to facilitate the initiation of a fatigue crack at a low-stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed:

70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

70.2.2 Method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three-point bending may be used to produce fatigue cracks.

Cantilever bending facilitates completely reversed loading

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(P_{min}/P_{max}= -1) which aids in starting and propagating the crack. For practical purposes, the K calibration for three-point bending may be used to estimate the stress-intensity factors for the cantilever bending.
NOTE: The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane that is normal to the notched face of the specimen.

70.2.3 Crack. The fatigue crack shall extend at least 0.04 in. beyond the tip of the crack starter notch. During at least the last 0.02 in. of fatigue cracking the following will apply:

(1) The ratio of the maximum-stress intensity in fatigue to Young's modulus, $K_{f(max)}/E$ shall not exceed 0.002 $in.^{1/2}$.

(2) The maximum stress intensity in fatigue shall be less than 60% of the stress intensity from the test, $K_{f(max)} < 0.6\ K_q$.

(3) The stress ratio P_{min}/P_{max} will fall between -1 and 0.1.

80. TEST PROCEDURE

80.1 Number of tests. Samples shall be selected as specified in the basic specification.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Fig. 2.

80.2.1 B and W. Measure the thickness, B, to the nearest 0.001 in. at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average value. Measure the depth W to the nearest 0.001 in. at two positions, one on each side of the crack starter notch and record the average value.

80.2.2 a. After fracture measure the initial notch plus fatigue crack length, a, to the nearest 0.5% at the following three positions: at the center of the crack front, a_3, and midway between the center and the intersection of the crack front with the specimen surfaces, a_2 and a_4. Use the average of these three measurements, $(a_2 + a_3 + a_4)/3$, as the crack length in the calculation of K_q (see 90.1). If the difference between any two of the crack length measurements exceeds 10% of the average, or if part of the crack front is closer to the machined notch root than 5% of the average, the specimen shall be discarded. Also, if the length of either surface trace of the crack, a_1 and a_5, is less than 80% of the average crack length, as defined above, the specimen shall be discarded. Record these five crack lengths.

80.3 Bend testing. Set up the bend-test fixture so that the

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line of action of the applied load shall pass midway between the support roll centers within 0.04 in. Measure the span (distance between the support-roll centers) within 1% and record this value as S. This value should fall between 1.50 in. and 1.65 in.

80.3.1 Location. Locate the specimen with the crack tip midway between the roll centers to within 0.040 in., and square to the roll axes within 2°. Check the parallelism between the top face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller shall be suitably adjusted.

80.3.2 Loading. Load the specimen at the rate between 700 lbf/minute and 3600 lbf/min. until the maximum load is reached, and record this value as P_{max}. Record the loading rate.

NOTE: It is desirable to follow the load change carefully because specimens having low toughness will break abruptly and no slow decrease in load beyond the maximum will be observed.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 K. Calculate a tentative fracture toughness value in KSI $\sqrt{in}$ from the measured value of P_{max} and the initial dimensions of the specimen as follows:

$$K = (P_{max}\ S/1000\ B\ W^{3/2})\ f(a/W)$$

90.2 Validity. Determine the validity of this tentative fracture toughness by checking the validity of the precracking (70.2.3), the shape of the crack front (80.2.2), and the bend test conditions (60.4, 80.3). If these requirements are met, this tentative value will be considered valid and will be reported as the fracture toughness, K_q, of the HF-1 steel.

90.3 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimen is given in Test Method E-399 and shall be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT

100.1 Report. The report shall include the following for each specimen tested:

100.1.1 All specimen dimensions (Section 80.2: B, W, a_1, a_2, a_3, a_4, a_5).

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100.1.2 All test parameters (Section 80.3: S, Load Rate, P_{max}, temperature).

100.1.3 A check of all test validators:

100.1.3.1 Precracking.

A. $-1 \leq (P_{min}/P_{max}) \leq 0-1$

B. $K_{f(max)}/E > 0.002$ in.$^{1/2}$

C. $K_{f(max)} < 0.6\ K_q$

100.1.3.2 Crack length.

A. $a_{fatigue} \geq 004$ in. + a_{notch}

B. $.45W < a < .55W$

C. a_1 and $a_5 > 0.8\ a$

D. $a_i - a_j < 0.1\ a$; where i,j = 2, 3, 4

100.1.3.3 Test conditions.

A. 1.5 in. < S < 1.65 in.

B. 700 lbf/minute < Load Rate <3600 lbf/min.

C. $T_{thermocouple} = T_{required} \pm 2°F$

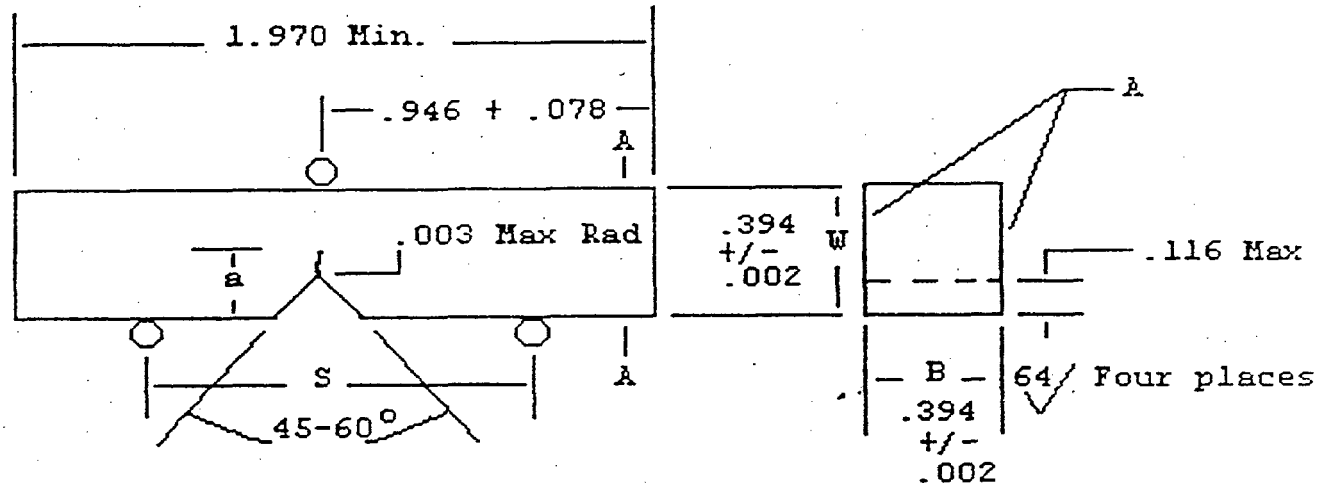
100.1.4 The fracture toughness, K_q, for all valid tests.

100.1.5 Crack plane orientation (ASTM E-399).

100.1.6 Fracture surface appearance (ASTM E-399).

100.1.7 0.2% offset tensile yield strength (ASTM E-8).

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NOTE 1. All dimensions in inches.

NOTE 2. "A" surfaces shall be perpendicular or parallel as applicable to within .001 in. TIR

NOTE 3. Crack starter shall be perpendicular to specimen length and thickness to within +/- 2 degrees.

FIGURE 2. Dimensions of precracked charpy specimens

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